

Machine Learning Image Classification for Agriculture Intelligence

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Introduction

The purpose of this research is to develop a machine learning model to classify and distinguish images of various crops when a limited amount of data is available.

Motivation

- Agricultural resilience requires intelligent monitoring of crop types
- Accurate fruit classification from images can help automate sorting, predict yield types, and support precision farming.

Objective

- Train a model that can accurately classify different crops.
- Use data augmentation and generative adversarial network (GAN) model to generate more training data.

Datasets

Agricultural Product Crop

Source: Kaggle Agricultural Crops Image Classification by MD Waquar Azam

Classification	Banana	Almond	Jute	Maize	Olives	Cherries
Binary	17, 7, 7	11, 5, 5	N/A	N/A	N/A	N/A
Multi-class	20, 6, 5	12, 5, 4	14, 5, 4	19, 5, 6	20, 4, 5	20, 6, 6

*(Images for training, Images for testing, Images for validation)

Methodology

Data Preparation

- Data augmentation is used to generate new and slightly different versions of the training set to increase the number of training images. These new images are randomly rotated, flipped, zoomed, and more.

Objective

- A baseline model using only the available training, validation, and testing images (very small dataset).
- Multi-class Feature Extraction program with Data Augmentation and Fine Tuning
- Binary-class Feature Extraction program with Data Augmentation, Fine Tuning, and Mis-labeling

Deep Learning Architectures

CNN model was used for binary and multiclass classification

- The CNN model has 4 layers with 1,215,262 neurons per sample

The pretrained model VGG16 was used for binary and multiclass classification

The pretrained model Xception was used for multiclass classification

Result and Discussion

Binary Models Accuracies

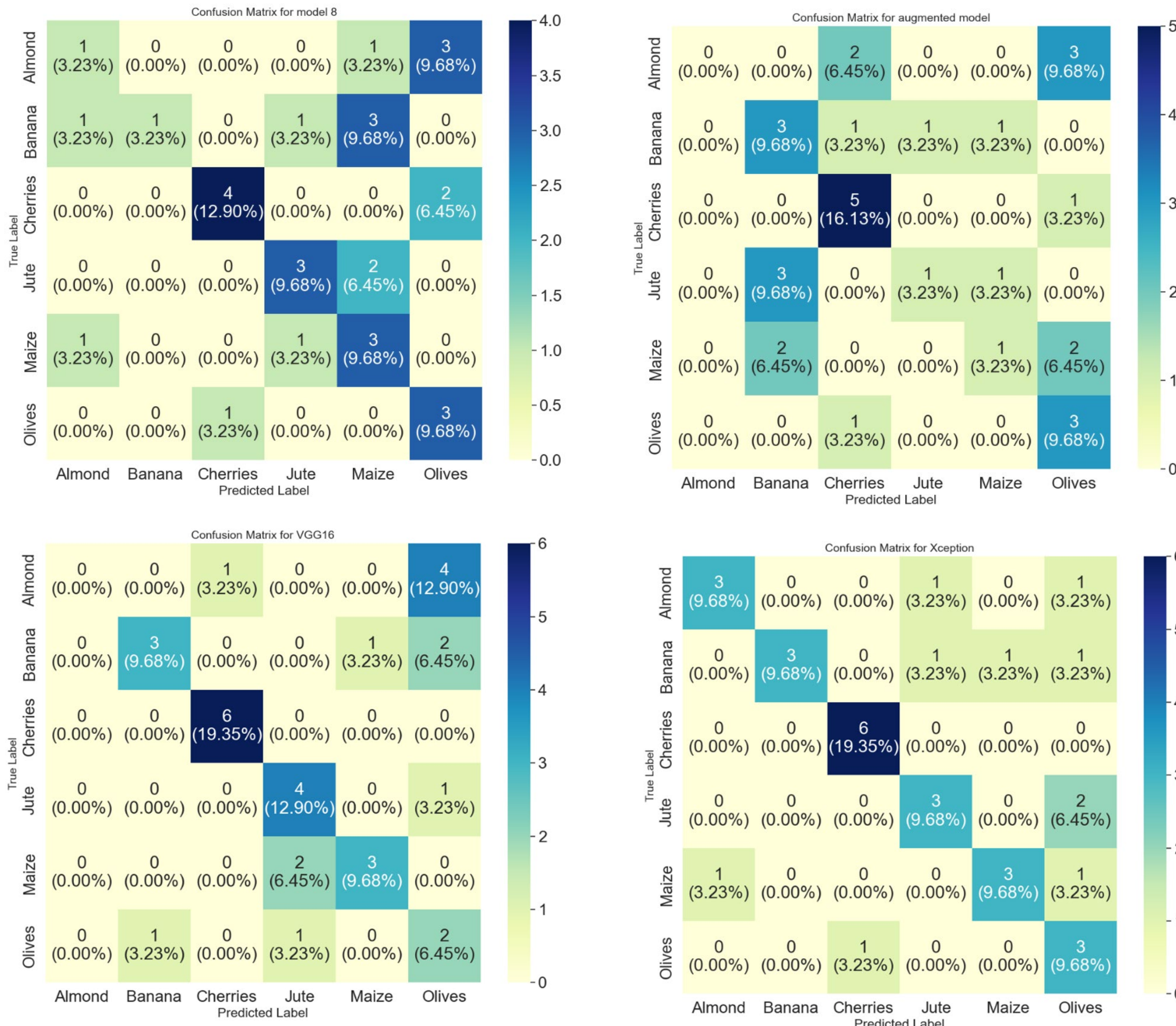
<i>Binary Model Features (Classes: Banana and Almond), all programs use the Pre-trained VGG16 Model</i>	<i>Mean Model Accuracy Based on 10 Saved Models</i>
Baseline Feature Extraction Model	78.33%
Feature Extraction Model using Fine Tuning	60.83%
Feature Extraction Model using Data Augmentation	70.00%
Feature Extraction Model using Fine Tuning and Data Augmentation	60.00%
Feature Extraction Model using Fine Tuning, Data Augmentation, and Mis-labeling	58.33%

*Mis-labeling: 5% of the training data for both classes (Banana and Almond) are images of a third class (Cherries)

Multiclass Models Accuracies

<i>Model Type</i>	<i>Accuracy</i>
Multiclass Baseline Model	48.39%
Multiclass Model with Data Augmentation	41.94%
Multiclass Model with Transfer Learning (VGG16) with Data Augmentation and Fine Tuning	67.74%
Multiclass Model with Transfer Learning (Xception) with Data Augmentation and Fine Tuning	74.19%

Multiclass Models Confusion Matrices (Baseline, Augmented, VGG16, Xception)



GAN Model

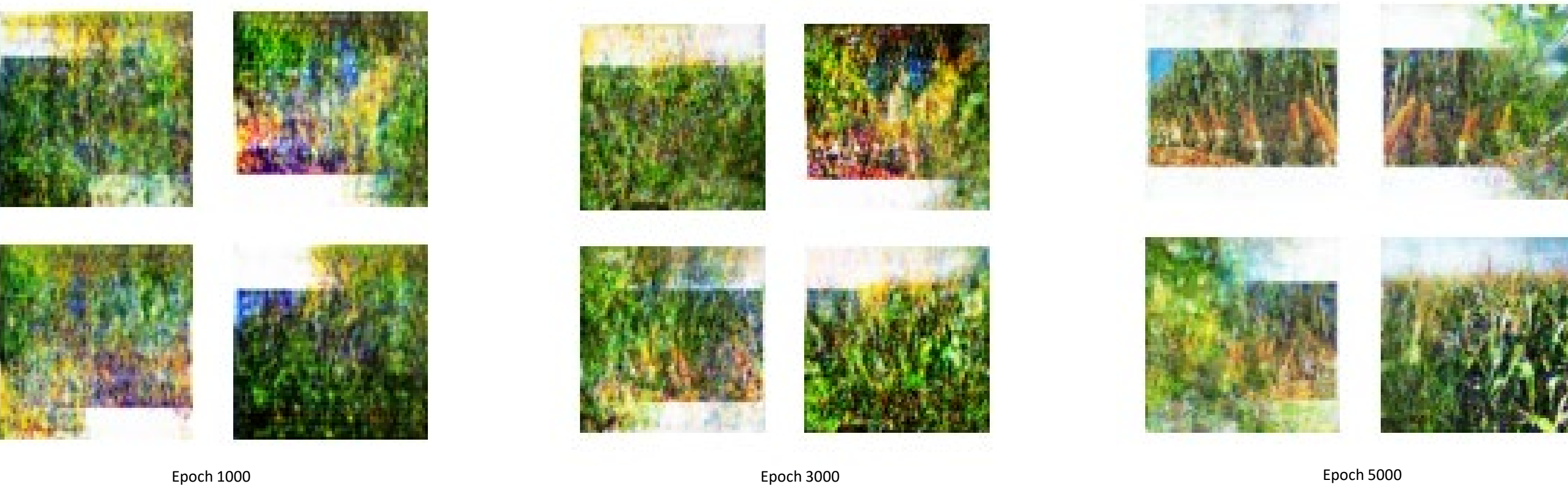
GAN Model

An attempt at generating new images with a GAN model was made. The results resembled the crops however, the images were not good enough to use as training images. Data Augmentation is used to get better quality image. The generated image size is 150 x 150 same as the training images. 19 images of maize is used to train the model. Normally it takes thousands of images to train a GAN Model. Therefore, we did not get expected image quality at the end. The model started to learn some features from 1000 epoch, and it improved a little bit later. It took around two and half hours to train the model with 5000 epoch.

GAN Model Architecture

- The generator model has 17 layers with 1,719,540 neurons per sample
- The discriminator model has 14 layers with 739,457 neurons per sample

Generated images of maize (Epoch 1000, 3000, 5000):



Conclusion

- Fine tuning and using a pretrained model improved the accuracies the most.
- For our models, the accuracy decreases when mislabeling is added.
- When using a GAN model for a small dataset, there needs to be several thousands of epochs for the generated images to resemble the training images.

